TEXT GENERATION USING GOOGLE GEMINI LLM API

Experiment / Practical Record

1. Aim

To develop a Python program using the Google Gemini Large Language Model (LLM) API to generate text responses for different user prompts and display the generated output in Google Colab.

2. Procedure

1. Open Google Colab and create a new notebook.
2. Install the required `google-genai` Python library.
3. Create an API key in Google AI Studio.
4. Run the program and enter the API key when prompted. The key is hidden while typing.
5. Initialize the Gemini client and select an available Gemini model.
6. Provide different prompts to the model using `generate_content()`.
7. Display each prompt and its generated response in the notebook output.

3. Python Code (Google Colab — Single Cell)

```
# Text Generation Using Google Gemini LLM API

!pip install -q google-genai

from google import genai
from getpass import getpass

# Enter your API key securely
api_key = getpass("Enter your Gemini API Key: ")

# Initialize Gemini client
client = genai.Client(api_key=api_key)
model = "gemini-2.5-flash"

# Different user prompts
prompts = [
    "Explain Artificial Intelligence in simple words.",
    "Write a short paragraph about Python programming.",
    "What are the benefits of renewable energy?"
]

# Generate and display responses
for prompt in prompts:
    print("\\nPrompt:", prompt)
    try:
        response = client.models.generate_content(
            model=model,
            contents=prompt
        )
        print("Generated Output:")
        print(response.text)
    except Exception as e:
        print("Error:", e)
    print("-" * 60)

print("\\nResult: Text generation using Google Gemini LLM API completed.")
```

4. Google Colab Output Screenshot (Illustrative)

The image below is a sample Colab-style output preview for the expected result. It is illustrative and is not a screenshot captured from a live Google Colab run; the actual responses may vary.

Google Colab

Sample output preview

Text Generation Using Google Gemini LLM API

```
!pip install -q google-genai
from google import genai
from getpass import getpass
api_key = getpass('Enter your Gemini API Key: ')
client = genai.Client(api_key=api_key)
model = 'gemini-2.5-flash'
# prompts sent to the Gemini model
```

Output

```
Prompt: Explain Artificial Intelligence in simple words.
Generated Output:
Artificial Intelligence (AI) enables computers to perform tasks
that normally require human intelligence, such as learning,
reasoning, and problem-solving.
------------------------------------------------------------
Prompt: Write a short paragraph about Python programming.
Generated Output:
Python is a popular programming language known for its simple
syntax and readability. It is used in web development, data
science, artificial intelligence, and automation.
------------------------------------------------------------
Prompt: What are the benefits of renewable energy?
Generated Output:
Renewable energy sources, such as solar and wind power, help
reduce pollution, decrease dependence on fossil fuels, and
support environmental sustainability.
------------------------------------------------------------
Result: Text generation using Google Gemini LLM API completed.
```

Illustrative sample only — actual Gemini responses may vary.

5. Result

Thus, a Python program was prepared to connect to the Google Gemini LLM API, send different user prompts, generate text responses, and display the output in Google Colab.

Notes

• Obtain an API key from Google AI Studio: https://aistudio.google.com/apikey
• Run the notebook in Google Colab: https://colab.research.google.com/
• Keep your API key private. If `gemini-2.5-flash` *is unavailable for your account, select a model currently supported by your Gemini API key.*